

when a patient's condition requires immediate care. These would happen in true real-time fashion, without any dependency on the cloud platform. At times, this could be a life-saver, especially in remote areas where connectivity can be unreliable.

- Edge computing devices are the main components of military equipment, according to Chaudhary. "Edge devices have business rules for different battle conditions. These edge devices acquire data in real time and respond as per the battlefield conditions. Furthermore, modern edge devices have larger storage capacities, allowing drones used in military surveillance to store more data for longer durations. This improved endurance enhances their effectiveness in extended operations," he says.

- Independent decision-making, without reliance on the cloud, is imperative for self-driving cars. The edge devices store AI models with details of different traffic conditions and the way the car must react to these. Autonomous vehicles use these to detect and drive through different traffic conditions. It would be impossible to wait for instructions to come from the cloud platform for every single situation, as delays are totally unacceptable in this use-case.

- Thanks to edge computing, drones have become invaluable in remote locations where communication is limited. "For instance, agricultural drones equipped with pesticide sprinklers can take videos and analyse local conditions of insect infections, allowing them to decide precisely where to sprinkle pesticides. This intelligent approach saves cost and effort, and reduces pesticide wastage, as the pesticide is targeted only to the affected areas," explains Chaudhary.

- Edge computing is a true ally in field operations, especially in remote locations with constrained resources. Take the example of elevator maintenance—elevators come in various

Benefits of Edge Computing

To the user:

- Low latency, i.e., faster response times
- More personalisation
- Real-time updates

To the business:

- Cost savings
- Bandwidth optimisation
- Decentralisation of data
- Real-time reports
- Enhanced security

models, making it difficult for a technician to be an expert in all of them, and connectivity is often disrupted in the lift maintenance room. In such scenarios, field technicians can receive support through augmented reality (AR) or virtual reality (VR) devices, which enable them to access large digital models, manuals, and an extensive knowledge database. Edge computing enables technicians to find such assistance even in remote operational sites, such as basements, where connectivity is limited.

- Chaudhary adds that remote diagnostics kits are also gaining popularity as they monitor the real-time status of machines, allowing for prompt diagnosis and error rectification as needed. "Advancements in edge computing facilitate closer collaboration with the command centre through video-based assistance, enabling quick issue resolution and avoiding any disruptions in operations," he says.

AI at the edge

Almost every use case we have discussed has AI involvement in it. Going a step ahead, experts feel that mobile devices can even run generative AI models, with up to a certain number of parameters. For example, LLaMa, a seven billion parameters AI model developed by the Stanford University, can run on a Snapdragon Insider smartphone.

It is believed that specialised AI models like, for example, an AI model for medical assistance with less than 100 billion parameters, will be more popular in the future than jacks-of-all-trades like ChatGPT. And these might well run on mobile devices. We are moving towards hybrid AI, where most of the execution will happen on the edge device (phones, tablets, computers, AR/VR headsets, etc), while training of the AI models will happen on the cloud.

Chaudhary sums up, "The integration of AI on the cloud and AI at the edge is of utmost importance in today's technological landscape. Cloud-based AI has revolutionised data processing and analysis with its vast computational capabilities and storage. However, edge AI complements this approach by offering unique advantages. It involves deploying AI algorithms and models directly on edge devices or gateways, closer to the data source, providing reduced latency, enhanced privacy and security, offline functionality, and bandwidth optimisation. By harnessing the strengths of both cloud-based AI and edge AI, industries can create a comprehensive and efficient AI ecosystem that addresses diverse use cases and ensures real-time decision-making and seamless operations."

According to a recent report by Grand View Research, the global edge computing market is projected to reach \$155.9 billion by 2030, growing at a compound annual growth rate (CAGR) of 37.9%. This is quite like the booming growth of cloud a couple of years ago. No wonder, edge is in the news almost every day—whether it is T-Mobile's initiative with Google Cloud's Distributed Cloud Edge, or SAP's collaboration with SingTel to develop 5G edge computing solutions. We can see a lot more interesting happenings in this space in the near future! Stay tuned. **EFY**